HOSPITAL CHATBOT

Split Voice System

Full Awareness Guide for Presenter

Everything we built · how it works · what to say

Read this before your PowerPoint talk

0. Quick map

1. What we built (definition)
2. Why we built it (problem)
3. Architecture (Planner + Executor)
4. Full pipeline (step by step)
5. VoicePlan (AI output contract)
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> **Core one-liner (memorize):**
> Whisper hears → GPT plans → Python proves against the hospital DB → WhatsApp replies with text/buttons.

1. What we built

The **Split Voice System** lets patients send a **WhatsApp voice note** to the hospital chatbot. The bot understands the speech and continues the **same booking / reschedule / cancel flow** used for typed messages.

Keywords: **Split Voice** **Router V2** **Planner** **Executor** **VoicePlan** **Whisper**

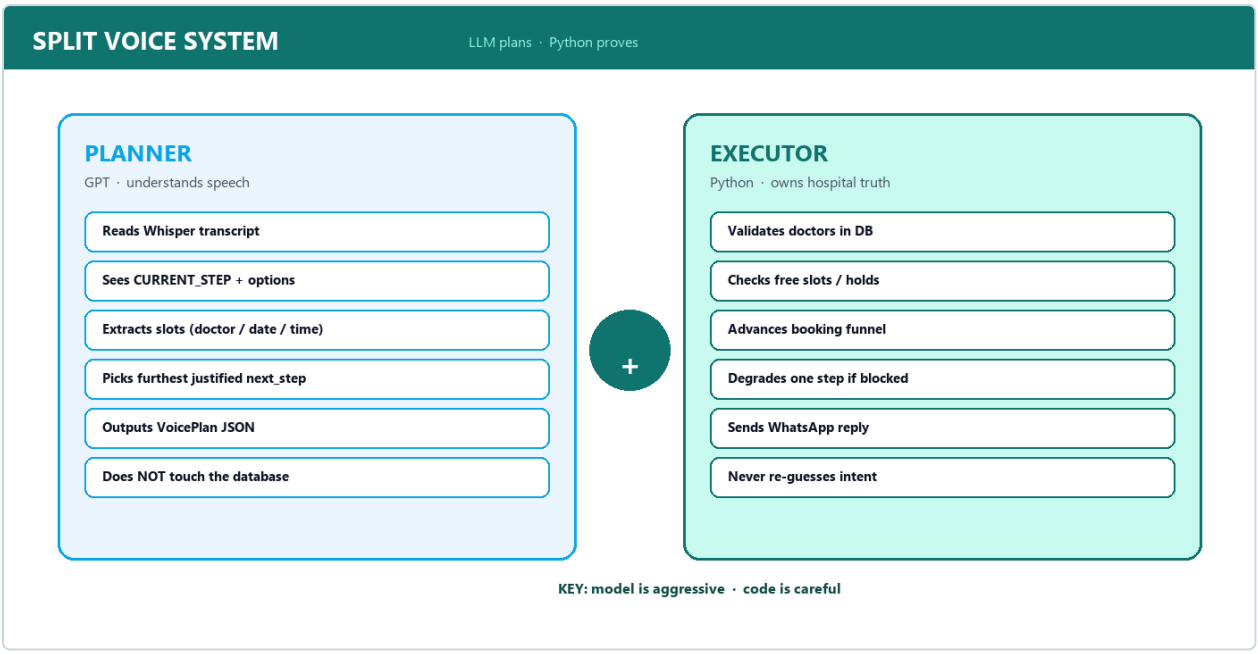
- Voice is an **alternate input**, not a separate product
- Replies are normal WhatsApp **text / buttons / lists**
- There is **no outbound TTS** (bot does not speak back)
- Hospital database is always the **source of truth**

2. Why we built it

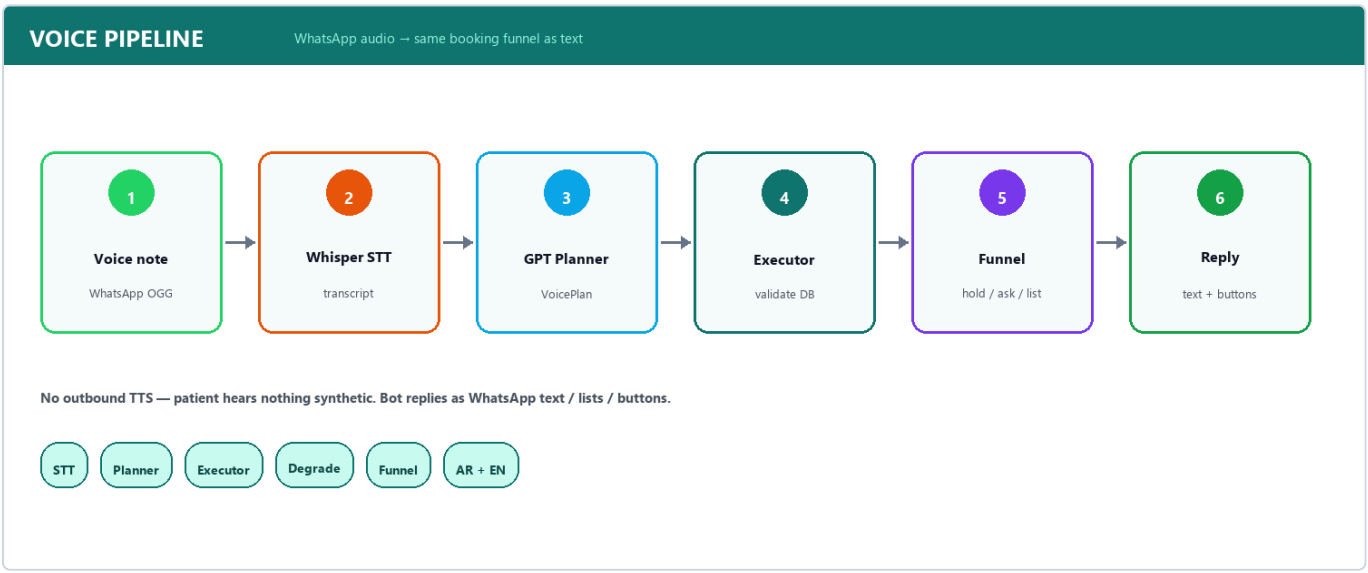
Problem	Solution
Typing is slow for many patients	Speak a voice note instead
Egyptian Arabic / mixed language hard to type	Whisper + dialect-aware planning
Long guided chats cause drop-off	Jump as far as the words justify
Old V1 voice routing hedged / duplicated menus	Clear Planner / Executor split (V2)

> **Say:** "We added voice to remove typing friction — without inventing a separate voice app."

3. Architecture — the split



Planner (AI) vs Executor (System)



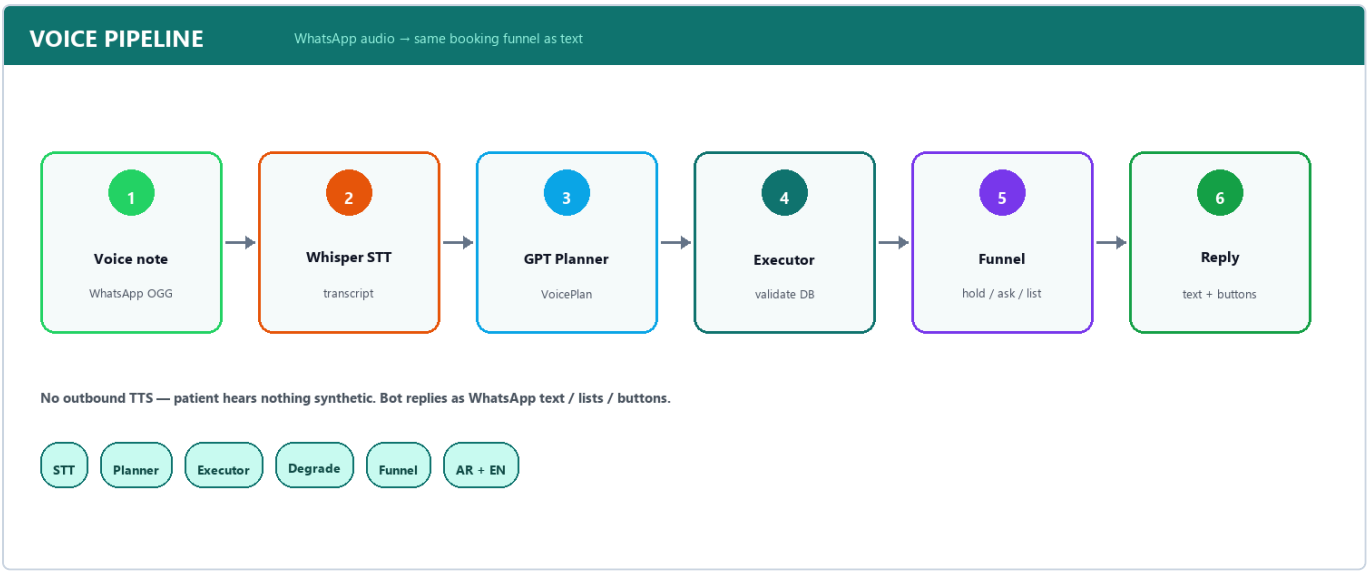
End-to-end voice pipeline

Layer	Owns	Does NOT own
Planner (GPT)	Understand transcript, extract slots, pick furthest justified next step, emit VoicePlan	Real doctors, free slots, DB writes
Executor (Python)	Validate vs PostgreSQL, hold slots, advance funnel, send WhatsApp reply, degrade safely	Re-guessing patient intent

> **Design rule:** LLM plans the furthest justified step. Python validates availability and falls back.

Production flag: `VOICE_ROUTER_V2=true` (default).

4. Full pipeline (what happens for one voice note)




```
1. Patient sends WhatsApp voice note (OGG audio)
2. Meta webhook → POST /webhook → message_type = audio
3. Background task: _handle_voice(phone, media_id)
4. Download audio from WhatsApp media API
5. Whisper STT → transcript (language NOT pinned — better Egyptian dialect)
6. Scrub empty / hallucination transcripts → ask clarify if needed
7. GPT planner → VoicePlan { goal, next_step, slots, confidence, language }
8. Save Message + VoiceMessage + cost usage log
9. LanguageService picks sticky reply language (AR/EN)
10. voice_executor.execute_plan:
      · check doctor / date / time against DB
      · hold slot OR show list OR ask next question
      · if blocked → degrade one step + explain
11. Patient sees WhatsApp text / buttons / lists
12. Same conversation_engine booking funnel as typed chat
```

Important: when a V2 plan is executed, we **do not** also run the old V1 fastpath — that was the duplicate-menu bug.

5. VoicePlan — what the AI returns

VoicePlan is the contract between AI and hospital logic.

Field	Meaning	Examples
`goal`	What patient wants	book_new, reschedule, cancel_appointment, abandon_draft, ask_doctors, handover, confirm, unclear…
`next_step`	Where funnel goes	ask_specialty, show_doctors, ask_date, show_slots, hold_slot, confirm_booking, pick_appointment, clarify…
`slots`	Extracted entities	doctor_name, specialty, date, time, change_target…
`confidence`	How sure the model is	per-field scores
`language`	Reply language hint	ar / en / mixed

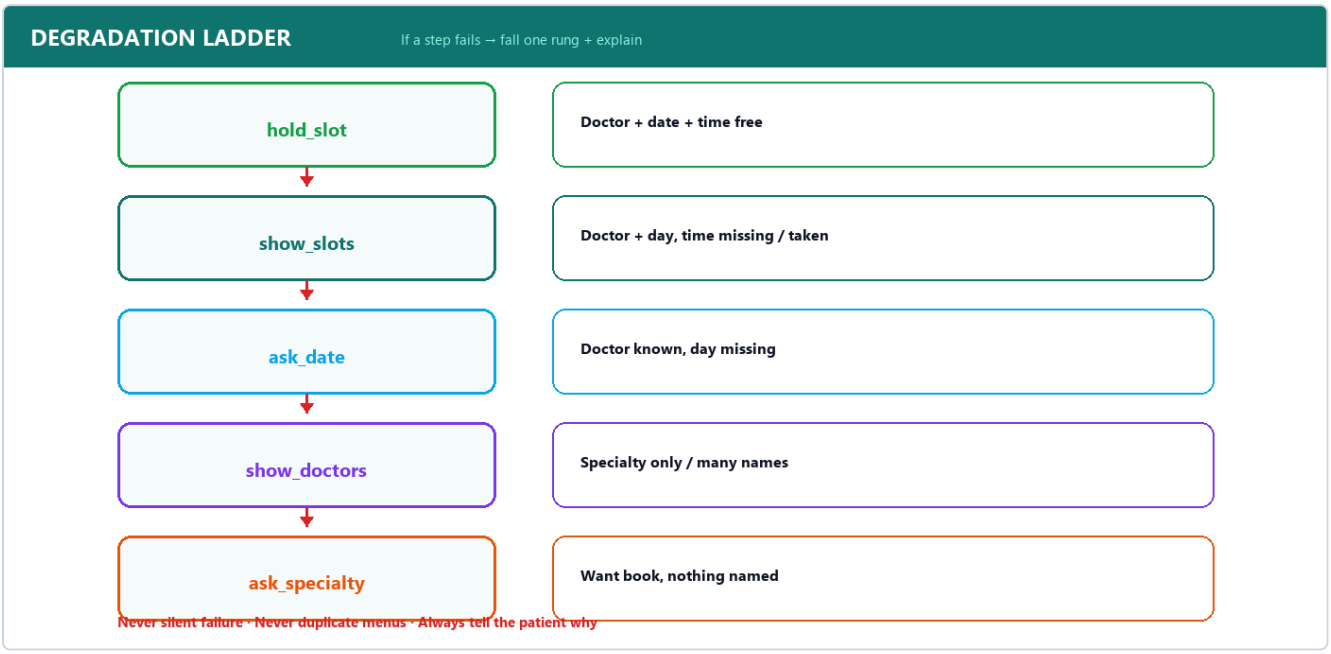
Step decision cheat sheet

```
doctor + date + specific time  → hold_slot
doctor + day, no time          → show_slots
doctor only                    → ask_date
specialty only                 → show_doctors
want book, nothing named       → ask_specialty
cancel booked appointment      → pick_appointment
abandon mid-draft              → show_menu
handover request               → handover
```

Important distinction

- `cancel_appointment` = cancel a **real booked** appointment
- `abandon_draft` = stop a half-finished booking ("خلاص / مش عايز")

6. Degradation / safety



If the planner asks for something unavailable, the executor falls **one rung** and explains.

```
hold_slot
   ↓ (time missing / taken)
show_slots
   ↓ (day missing)
ask_date
   ↓ (doctor unclear)
show_doctors / ask_specialty
```

- Never invent a doctor or free slot
- Never silent failure
- Never send two conflicting replies for one voice note

> **Say:** "AI can be aggressive. Code is careful. If 2pm is taken, we show open slots and say so."

7. Languages

- Whisper is **not language-pinned** (better Egyptian dialect quality)
- Optional STT vocab prompt is **OFF** by default (`VOICE_STT_PROMPT=false`)
- Reply language chosen **after** STT from transcript script (Arabic vs Latin)
- Sticky language window: `VOICE_LANG_SESSION_MINUTES` (default 30)
- Supports Egyptian Arabic, English, and code-switch
- Bot replies in one language per message (AR or EN), not mixed

8. V1 vs V2

V1 (legacy)	V2 (production)
Model picked ~5 rough actions	Model emits full VoicePlan
Python re-decided with thresholds	Python only validates + runs
Hedging + duplicate menus	Furthest step + one-rung degrade
Fastpath maze mid-flow	No silent fall-through when plan exists
Hard to reason about	Clear Planner / Executor split

Rollback: set `VOICE_ROUTER_V2=false`.

9. Code modules & config

File	Role
`webhook.py`	Detect audio, queue `_handle_voice`
`voice_service.py`	Download OGG, Whisper STT, GPT plan entry
`voice_prompts.py`	Static `PLANNER_SYSTEM` (prompt-cache friendly)
`voice_schemas.py`	`VoicePlan` / `VoiceSlots` types
`voice_executor.py`	`execute_plan` — validate, degrade, reply
`voice_fastpath.py`	V1 mid-flow only (fallback)
`voice_cost.py`	STT + GPT cost logging
`language_service.py`	Post-STT reply language
`conversation_engine.py`	Shared booking state machine (text + voice)

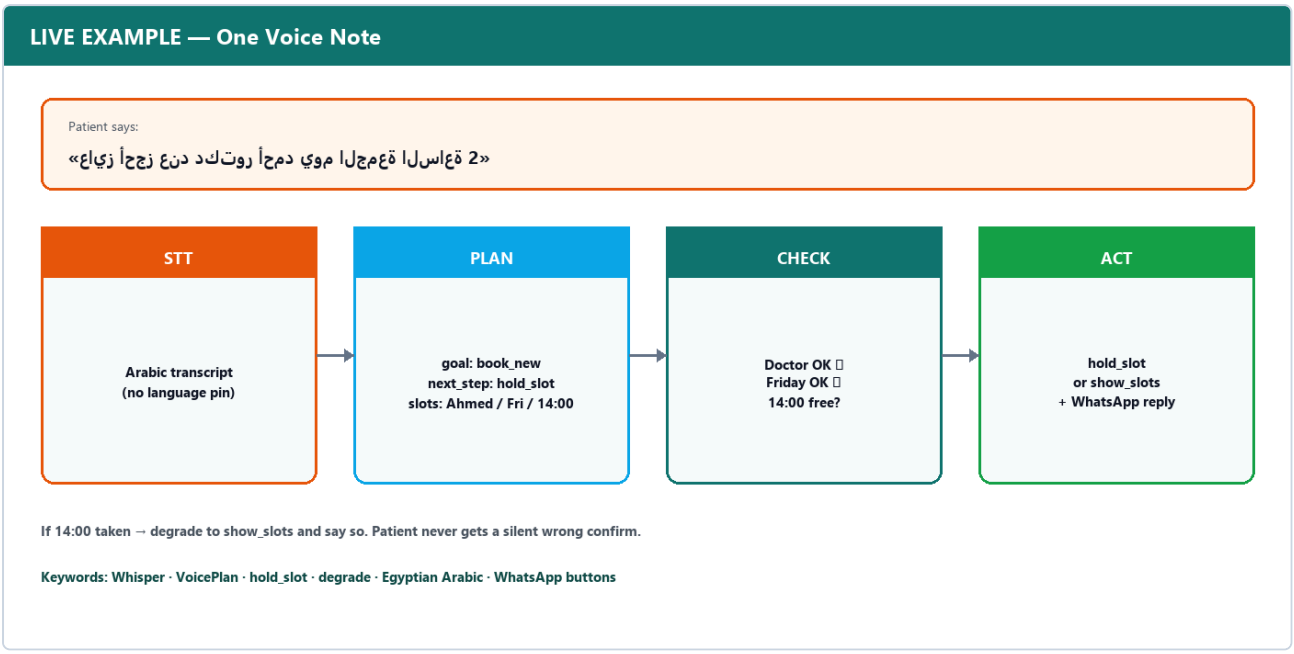
Config knobs

Env	Default	Meaning
`OPENAI_STT_MODEL`	whisper-1	Speech-to-text
`OPENAI_VOICE_STRUCTURE_MODEL`	gpt-4o-mini-ish	Planner model
`VOICE_ROUTER_V2`	true	Split planner/executor path
`VOICE_STT_PROMPT`	false	Extra Whisper vocab prompt
`VOICE_LANG_SESSION_MINUTES`	30	Sticky reply language
`VOICE_REQUIRE_CONFIRM`	false	Extra voice confirm gate

10. Admin / cost / observability

- Transcripts stored on `VoiceMessage`
- Usage logs track voice STT + voice parse cost
- Streamlit dashboard KPIs: voice notes, minutes, cost, avg $/voice
- Tests: `test_voice_router_v2.py`, state-aware, language, cost, date/time hints…
- Eval helpers at repo root: `eval_voice_*.py`

11. Example walkthrough



Demo story:
Patient says: «عايز أحجز عند دكتور أحمد يوم الجمعة الساعة 2»
→ Whisper transcript
→ Plan: goal=`book_new`, next_step=`hold_slot`, slots=Ahmed / Friday / 14:00
→ Executor checks DB
→ If free: hold + ask confirm
→ If not: degrade to `show_slots` and explain

12. What is NOT included

- No outbound text-to-speech / robot voice replies
- No Azure Speech / ElevenLabs
- No separate voice mobile app UI (WhatsApp native recorder only)
- `edge-tts` / `gTTS` may appear in requirements but are not used by this voice path

13. Talk track for your PPT (4 slides)

Your PPT slide	What you say
Keywords / title	"Patients can book by WhatsApp voice note. Voice in, buttons out."
Why Voice	"Typing friction + dialect. Voice removes friction without a separate app."
Architecture	"Whisper → Planner AI → Executor checks DB → WhatsApp reply. AI suggests, system confirms."
Key points	"Input only · AI plans · system proves · no robot voice. Same product as typed chat."

30-second version

> Patients send WhatsApp voice notes. Whisper turns speech into text. GPT makes a booking plan. Python checks the real hospital schedule. If the time is free we hold it; if not we step back and explain. Replies are normal WhatsApp buttons. Safe, Arabic + English, same booking product.

Q&A cheat sheet

Question	Answer
Do you reply with voice?	No — WhatsApp text/buttons only.
What converts speech?	OpenAI Whisper (`whisper-1`).
Who decides next step?	AI Planner suggests · Executor confirms with DB.
Can AI invent a doctor/time?	No. Executor validates everything.
What if time is taken?	Degrade one step (e.g. show_slots) and explain.
Arabic support?	Yes — Egyptian Arabic + English + code-switch.
Is voice a separate bot?	No — same conversation funnel as typing.
Can we roll back?	Yes — `VOICE_ROUTER_V2=false` .
Where is cost tracked?	Usage logs + Streamlit voice KPIs.

> **Bottom line:** Split Voice = understand speech aggressively, book only what the hospital DB proves is real.